

The Negative And Complex Extensions Of The Fibonacci Sequence Using Binet's Formula

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Abstract

Binet's Formula, which produces the n th value in the Fibonacci Sequence can be used to extend the traditional series in the negative direction, between integer sequence positions, and into the complex plane.

1 Introduction

Named from the French Mathematician, Jacques Philippe Marie Binet, Binet's Formula, shown below, calculates the n th number within the Fibonacci Sequence. The Fibonacci Sequence begins with a 0 at the 0th position as well as a 1 at the first position, and derives subsequent integer terms from the addition of the prior two.

- The Fibonacci Sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...
- Binet's Formula: *Where n is the position of the n th Fibonacci Number*

$$\frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

2 Negative Inputs

2.1 Without Binet's Formula

Even without Binet's Formula, one can calculate the preceding term, x_{n-1} , by using the following formula: $x_{n-1} = x_{n+1} - x_n$. As such,

$$\begin{aligned} 1 - 0 &= 1 \text{ (This is the -1st term)} \\ 0 - 1 &= -1 \text{ (The -2nd term)} \\ 1 - (-1) &= 2 \text{ (The -3rd term)} \end{aligned}$$

...

2.2 The Negative Fibonacci Numbers

By simply inputting the negative integers into Binet's Formula, or by back-calculating the series (2.1), the extended Fibonacci Sequence is shown below:

... -55, 34, -21, 13, -8, 5, -3, 2, -1, 1, 0 (The 0th Term), 1, 1, 2, 3, 5, 8, 13, 21, 34, 55 ...

With alternating negative/positive signs on the negative portion of the sequence preserving the $x_{n-1} + x_n = x_{n+1}$ property.

3 The Complex Spiral Between The Fibonacci Numbers

3.1 Python Calculations

Due to the fact that the $\frac{1-\sqrt{5}}{2}$ term evaluates to a negative number, non-integer powers of n produce complex values, and as such require a more specific analysis. The coding language Python 3 was selected and used to process decimal n -values using the following code:

```
import cmath
import math
n=-10
while n<10:
    f=(1/math.sqrt(5))*(((1+math.sqrt(5))/2)**n-((1-math.sqrt(5))/2)**n)
    print(f.real,f.imag)
    n=n+0.1
```

Do note the $n = -10$ and $n < 10$ terms control the length of the series, from -10 to 10, and the $n = n + 0.1$ line represents the increments/precision, +0.1, of the code. The output of the code, a list of the real and imaginary components of the Fibonacci Numbers, were then placed in order pairs in the x-y complex plane.

3.2 The Plots

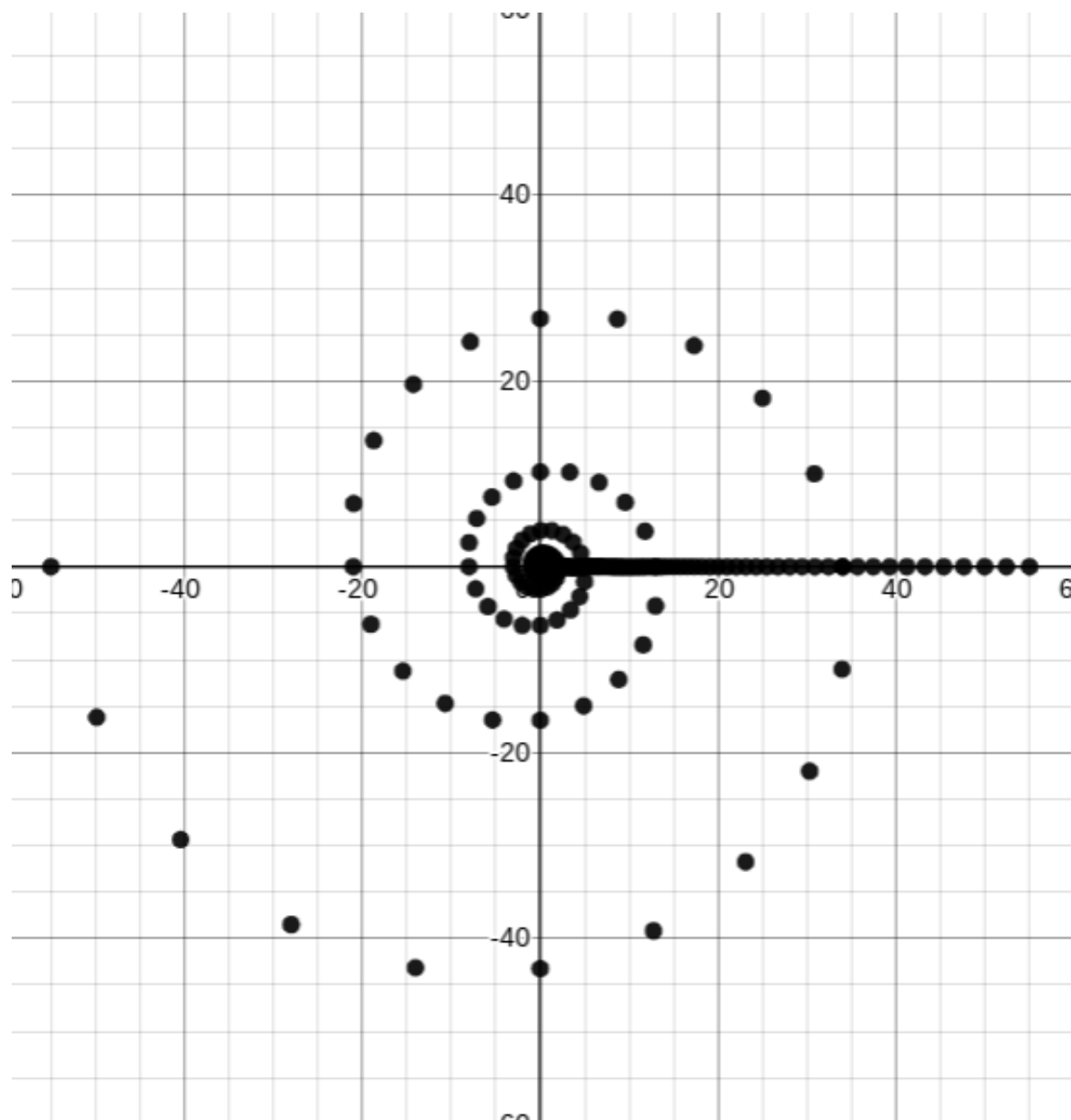


Figure 1: The Marvelous Complex Fibonacci Spiral.

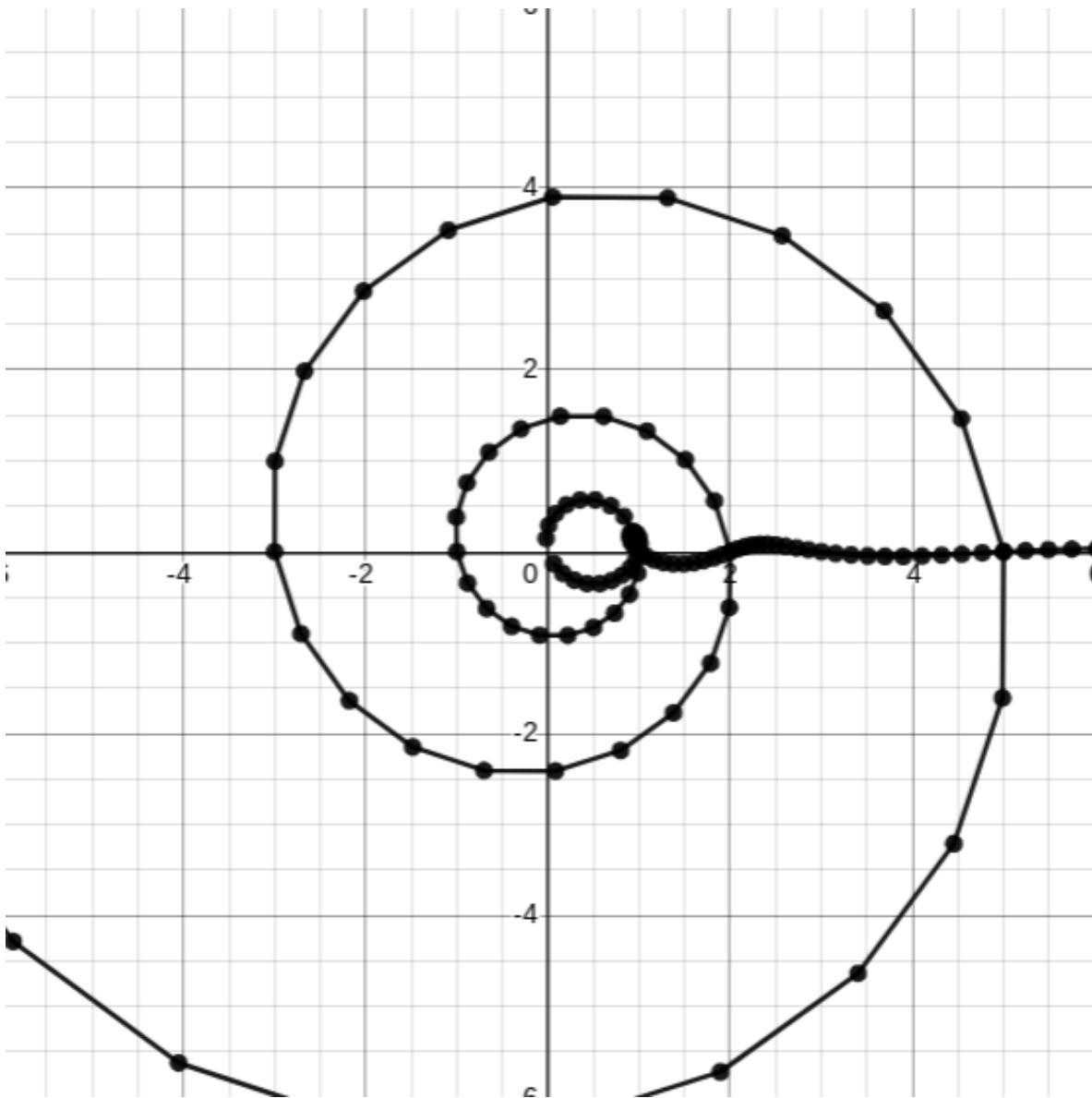


Figure 2: Adjacent Points Are Connected By Lines

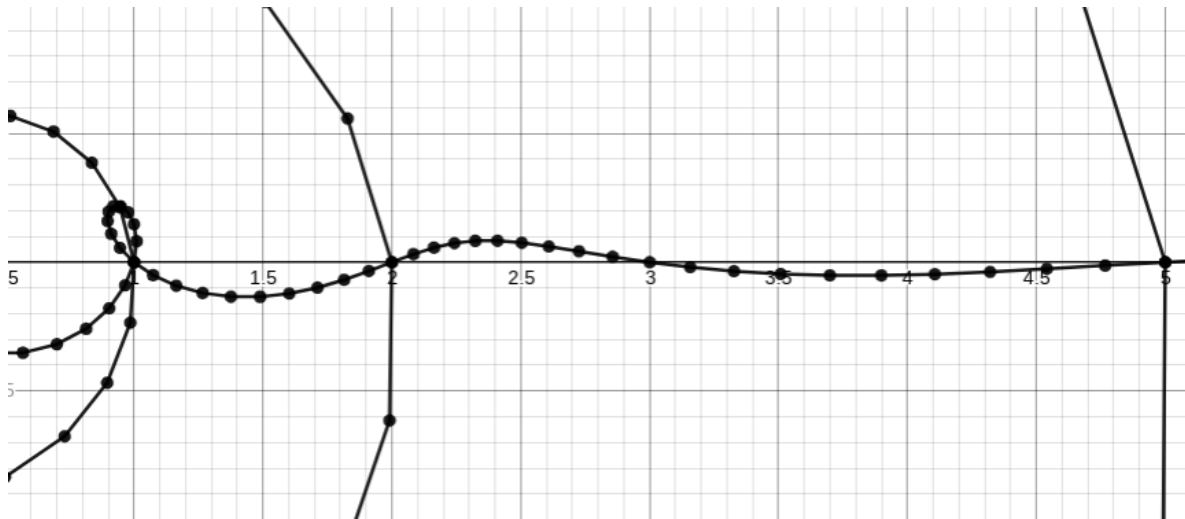


Figure 3: Centered On X-Axis

4 Personal Observations

- The Fibonacci Sequence appears to be only a part of a larger construct, at the x-axis integer intersections, specifically.
- A gradually flattening sinusoid-like curve crosses the positive x-axis at every Fibonacci Number.
- Since the positive terms of the negative side of the Fibonacci Sequence (2) overlap with the identical positive terms, the spiraling arms intersect the positive x-axis every-other 'zero' of the gradually flattening sinusoid-like curve. This has the added effect of making the negative x-axis appear half as populated.
- Visible within Figure 3, observe the double-looping, triple-intersection effect near the $x=1$ region, as the negative-hand of the Fibonacci Series (2) reaches 1, and the positive-hand series repeats 1 and loops back to intersect 1 once more.